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4.2 Organisation

AQA GCSE Biology (8461) — Paper 1

Multicellular organisms are organised into levels:
cell → tissue → organ → organ system → organism.
A **tissue** is a group of similar cells working together; an **organ** is a group of different tissues working together; an **organ system** is multiple organs working together.

4.2.1 Principles of Organisation

Tissue	Function	Example organ
Muscular	Contraction for movement	Heart, stomach, skeletal muscle
Glandular	Produces & secretes substances (enzymes/hormones)	Pancreas, salivary glands
Epithelial	Covers surfaces — protection, absorption, secretion	Lining of small intestine
Nervous	Transmits electrical impulses	Brain, spinal cord

4.2.2 Enzymes and Digestion

Enzymes

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Enzymes are biological catalysts — they speed up chemical reactions in the body without being used up. They are made of **protein**.
Each enzyme has a specific **active site** whose shape is complementary to its substrate (**lock and key model**).

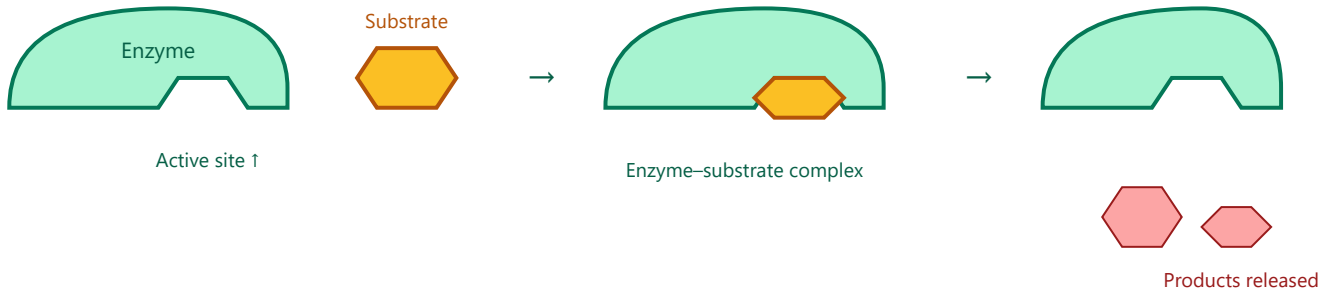

Fig 1: The lock-and-key model — the substrate fits exactly into the active site; the products are released, and the enzyme is unchanged.

Effect of Temperature and pH on Enzymes

Temperature: Rate increases with temperature up to an **optimum**. Above the optimum, the bonds holding the active site break; the enzyme is **denatured** and rate drops sharply.

pH: Each enzyme works best at a specific pH. Extreme pH causes denaturation.

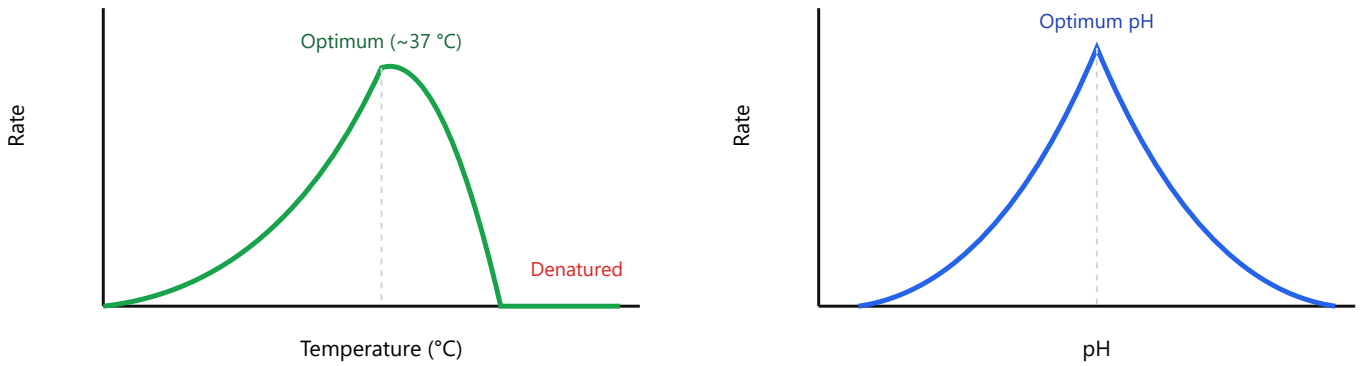

Fig 2: Left — rate of enzyme reaction vs temperature. Right — rate vs pH. Both show clear optimum points.

The Digestive System

Enzyme	Where produced	Substrate → Products	Optimum pH
Amylase (carbohydrase)	Salivary glands, pancreas	Starch → maltose	~7
Maltase	Small intestine wall	Maltose → glucose	~7
Protease (pepsin)	Stomach	Protein → amino acids	~2
Protease (trypsin)	Pancreas, small intestine	Protein → amino acids	~8
Lipase	Pancreas, small intestine	Lipids → fatty acids + glycerol	~7.5

Bile

Bile is made in the **liver**, stored in the **gallbladder** and released into the small intestine. It has two functions:

- **Neutralises acid** from the stomach — providing optimum pH for small intestine enzymes.
- **Emulsifies fats** — breaks large fat droplets into small ones, increasing surface area for lipase.

Food Tests

Test for	Reagent	Method	Positive result
Starch	Iodine solution	Add a few drops to sample	Yellow/brown → blue-black
Reducing sugar	Benedict's	Heat with sample at 60–80 °C	Blue → brick-red precipitate
Protein	Biuret	Add in alkaline conditions	Blue → purple/violet
Lipids	Ethanol/Sudan III	Add ethanol, shake with water (or Sudan III)	Cloudy white emulsion / red staining

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4.2.3 The Heart and Blood Vessels

Humans have a **double circulatory system**: blood passes through the heart twice per circuit — once to the lungs (pulmonary) and once to the body (systemic).

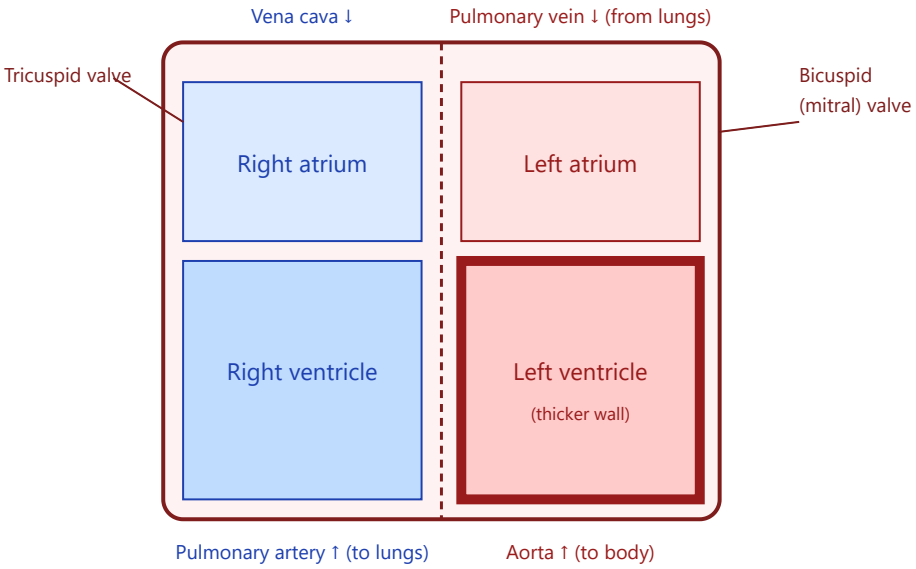

Fig 3: Human heart — the left ventricle has a thicker wall because it pumps blood at high pressure around the body.

Pathway of Blood Through the Heart

Deoxygenated blood: **Vena cava → Right atrium → Right ventricle → Pulmonary artery → Lungs.**
Oxygenated blood: **Pulmonary vein → Left atrium → Left ventricle → Aorta → Body.**

Blood Vessels

Feature	Artery	Vein	Capillary
Wall	Thick, muscular & elastic	Thinner walls	One cell thick
Lumen	Narrow	Wide	Very narrow
Pressure	High	Low	Low

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Valves?	No	Yes (prevent backflow)	No
Direction	Away from heart	Toward heart	Exchange with tissues

Blood Components

- **Red blood cells** — contain haemoglobin which binds O_2. Biconcave, no nucleus — maximises surface area and space for haemoglobin.
- **White blood cells** — defend the body. *Phagocytes* engulf pathogens; *lymphocytes* produce antibodies.
- **Platelets** — small cell fragments that produce clots to seal wounds.
- **Plasma** — yellow liquid carrying cells, nutrients, hormones, urea, CO_2 and heat.

Heart Rate Equation

Cardiac output (cm^3/min) = stroke volume (cm^3) × heart rate (bpm)
Example: stroke volume 70 cm^3, heart rate 75 bpm → 70 × 75 = 5250 cm^3/min.

Coronary Heart Disease (CHD) & Treatments

CHD occurs when fatty deposits (atheroma) build up in the **coronary arteries** that supply the heart muscle, reducing blood flow. It can lead to angina or a heart attack.

Treatment	How it works	Advantages	Disadvantages
Statins	Drugs lowering LDL cholesterol	Non-invasive; reduce CHD risk	Daily use; side effects (muscle pain, liver)
Stents	Mesh tubes open narrowed arteries	Immediate improvement of blood flow; long-lasting	Surgical risk; possible re-narrowing
Heart transplant	Replace damaged heart with donor	Cures severe failure	Donor shortage; risk of rejection; lifelong medication

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Mechanical/artificial heart	Pump supports or replaces heart	No rejection; bridge to transplant	Limited lifespan; clot risk
Biological vs mechanical valve	Replace faulty valve	Biological — no anticoagulants needed	Biological — wears out quicker; mechanical — lifelong anticoagulants

4.2.4 Non-Communicable Disease

Non-communicable diseases cannot spread between people. **Risk factors** are things that raise your chance of disease. Risk factors for CHD: *smoking, high-fat diet, obesity, inactivity, genetics, high alcohol intake*.

Cancer

- **Benign tumour** — grows in one place and does not invade other tissues (usually not life-threatening).
- **Malignant tumour** — cells break off and spread in the blood to form secondary tumours (metastasis) — cancerous.

Lifestyle risk factors: smoking, UV exposure, alcohol, poor diet, obesity. Genetic risk factors also exist (e.g. BRCA1/2 genes for breast/ovarian cancer).

4.2.5 Plant Tissues, Organs and Systems

Leaf Structure

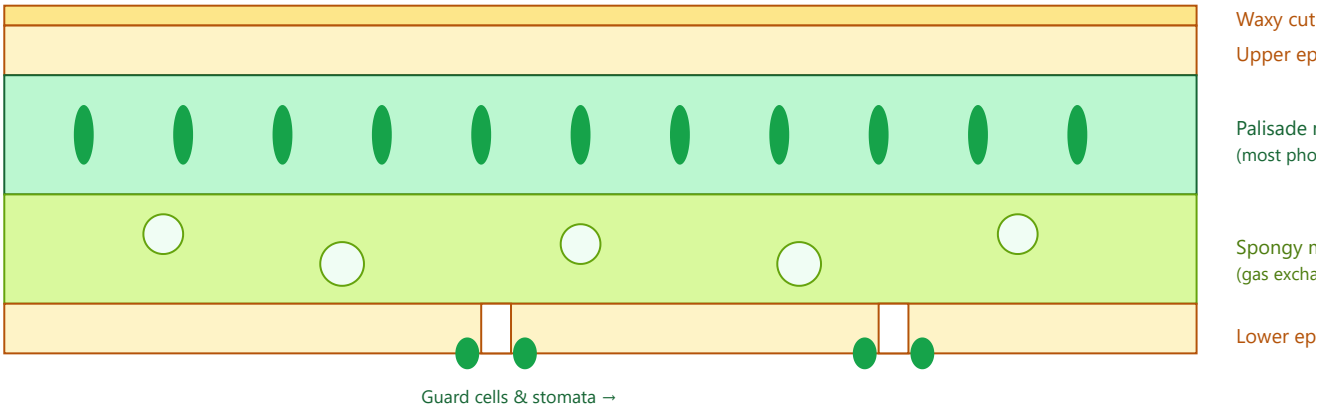

Fig 4: Cross-section of a leaf — palisade mesophyll is packed with chloroplasts; stomata on the lower surface allow gas exchange.

Transport Vessels: Xylem and Phloem

Vessel	What it carries	Direction	Process
Xylem	Water & mineral ions	Roots → leaves	Transpiration stream (driven by evaporation)
Phloem	Dissolved sugars (sucrose)	Leaves ↔ all parts (source → sink)	Translocation (active process)

Transpiration

> **Transpiration** is the loss of water vapour from plant leaves. Water evaporates from mesophyll cells and diffuses out through the stomata. This pulls more water up the xylem from the roots.

Factors that **increase** transpiration rate:

- Higher temperature (faster evaporation)
- Higher light intensity (stomata more open)
- More air movement/wind (removes water vapour)
- Lower humidity (steeper gradient)

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Rate of water uptake can be measured with a **potometer**. The distance an air bubble moves in a set time gives the rate.

Stomata and Guard Cells

Stomata are openings, mainly on the underside of leaves. **Guard cells** flank each pore:

- When turgid (full of water) — stomata **open** to allow gas exchange.
- When flaccid — stomata **close**, reducing water loss during drought.

> **Exam Tip:** A classic 6-mark question is: "Describe and explain how a leaf is adapted for efficient photosynthesis." Structure your answer by naming each tissue AND giving the adaptation (e.g. "Palisade mesophyll cells are column-shaped and packed with chloroplasts to maximise light absorption").

Summary — 4.2

- Organisation: cells → tissues → organs → systems → organism.
- Enzymes follow lock-and-key; activity peaks at optimum T and pH; denature when bonds break.
- Digestive enzymes: amylase (starch), protease (protein), lipase (fat).
- Bile emulsifies fat and neutralises stomach acid.
- Heart has 4 chambers; left ventricle is thicker (pumps to body).
- Arteries — thick muscular walls, high pressure; veins — valves, low pressure; capillaries — 1-cell-thick walls for exchange.
- Leaf: palisade (light) + spongy (gas exchange) + stomata; xylem up, phloem around.
- CHD treated with statins, stents, transplants or artificial hearts.

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